

MINIMAL FOLIATIONS, CODIMENSION-ONE STABLE NORMS, AND A QUESTION OF BANGERT

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ABSTRACT. We compute the codimension-one stable norm for a natural class of cohomogeneity-one metrics on tori. In every dimension $n \geq 3$, the formula yields smooth nonflat metrics for which each primitive codimension-one homology class is represented by a foliation of calibrated tori, giving a negative answer to a question of Bangert. On $\mathbb{T}^3$, we construct an infinite-dimensional family of nonflat metrics whose codimension-one stable norm agrees exactly with that of the unit cubic flat torus and whose total volume is fixed. An explicit two-parameter subfamily contains pairwise non-isometric metrics. These examples also show that the Euclidean-stable-norm-and-volume data are not locally injective near the cubic flat metric.

1. INTRODUCTION

In his contribution to the 1994 International Congress of Mathematicians, Bangert asked whether an abundance of codimension-one minimizing hypersurfaces characterizes flat Riemannian tori [4]. More precisely, he posed the following question.

Problem 1.1. Suppose that $(\mathbb{T}^n, g)$ is a Riemannian torus such that for every primitive class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, there exists a foliation of $\mathbb{T}^n$ by g -minimal $(n-1)$ -tori in the class r . Must g be a flat metric?

The motivation comes from the geodesic case on $\mathbb{T}^2$, where the answer is positive. Suppose that for every primitive class in $H_1(\mathbb{T}^2, \mathbb{Z})$, the torus is foliated by closed geodesics representing that class. Each such foliation lifts to a foliation of $\mathbb{R}^2$ by globally minimizing geodesics. Innami [12] proved that this condition forces the metric to be flat. This is closely related to Hopf's theorem [11] that a Riemannian two-torus without conjugate points is flat. The corresponding rigidity theorem in every dimension was proved by Burago and Ivanov [5].

For hypersurfaces in higher-dimensional $\mathbb{T}^n$, minimal hypersurfaces in such a foliation are still homologically area-minimizing. This is because any such foliation is calibrated: if ν is its global unit normal, then

$$d(\iota_\nu dV_g) = (\operatorname{div}_g \nu) dV_g = 0,$$

hence $\iota_\nu dV_g$ is a calibration form on every leaf. Bangert's hypothesis can therefore be stated equivalently using foliations by homologically area-minimizing tori.

Moser initiated a higher-dimensional Aubry-Mather theory for periodic variational problems and proved a stability theorem for certain minimal foliations [15, 16]. Bangert developed the corresponding structure theory of non-selfintersecting minimizers, introducing secondary invariants and describing the resulting foliations and laminations with gaps on the torus [2].

The natural invariant associated with these minimizing hypersurfaces is the codimension-one stable norm. For a closed oriented Riemannian n -manifold (M, g) and a class $r \in H_{n-1}(M, \mathbb{R})$, its stable norm is defined by the minimum mass of a real $(n-1)$ -cycle representing r . This mass norm on real homology goes back to Federer [7]. In codimension-one, it may be regarded as the effective or homogenized area functional associated with the Riemannian metric. A calibrated hypersurface realizes the stable norm of its homology class.

In a recent work [17], Nguyen considers, for an integer class $r \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, the min-max excess quantity:

$$\Delta W_r = \omega(r) - S(r),$$

where $\omega(r)$ is the Almgren-Pitts width defined using sweep-out inside r and $S(r)$ is the stable norm. He proves that for a totally irrational class α , $\Delta W_r \rightarrow 0$ as $r/S(r) \rightarrow \alpha/S(\alpha)$ if and only if there is a foliation by minimizing hypersurfaces in the direction α . Throughout this paper, we will use the more standard notation, $\|r\|_{\text{st}}$, to denote the stable norm.

The regularity of the stable norm is closely connected with the geometry of minimizing laminations. Auer and Bangert [1] proved that, at a nonzero class α , the restriction of the codimension-one stable norm to the smallest rational subspace containing α is differentiable. Junginger-Gestrich related differentiability in the remaining directions to the absence of gaps in the corresponding minimizing lamination [13]. Combining these results with a compactness argument for rationally dependent classes, we prove in Appendix A that, on a torus, Bangert's condition is equivalent to the fact that the stable norm on $H_{n-1}(\mathbb{T}^n, \mathbb{R})$ is differentiable at every nonzero class. Thus Bangert's problem can also be formulated as a rigidity question for the stable norm:

$$\|\cdot\|_{\text{st},g} \text{ is differentiable on } H_{n-1}(\mathbb{T}^n, \mathbb{R}) \setminus \{0\} \stackrel{?}{\implies} g \text{ is flat.}$$

Bangert [3] constructed a metric on $\mathbb{T}^n$ for which all but one primitive codimension-one homology class admit foliations by closed minimal hypersurfaces (see also [17]). The construction below is motivated in part by this example and gives a negative answer to Bangert's question.

1.1. Main results. Consider for $n \geq 3$ the torus

$$\mathbb{T}^n = \mathbb{T}_x^{n-1} \times S_t^1$$

with local coordinates $(x_1, \dots, x_{n-1}, t)$ and consider Riemannian metrics of the form

$$g = q(t)^2 dt^2 + h_t, \quad \det h_t = 1, \quad (1.1)$$

where $q : S^1 \rightarrow (0, \infty)$ is smooth and h_t is a smooth periodic family of translationally invariant metrics on $\mathbb{T}^{n-1}$. Geometrically, $t \mapsto h_t$ is a loop of marked flat, unit-volume $(n-1)$ -tori. We will see that the determinant condition preserves the volume of the horizontal fibers, making them minimal with respect to the metric g .

Theorem 1.2. *Let g be a metric of the form (1.1). For $(z, c) \in \mathbb{R}^{n-1} \times \mathbb{R}$, consider the homology class $r_{z,c} \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ defined by the Poincaré dual:*

$$\text{PD}(r_{z,c}) = \sum_{i=1}^{n-1} z_i [dx^i] + c[dt].$$

Its stable norm is given by

$$\|r_{z,c}\|_{\text{st},g} = \sqrt{\mathcal{A}_g(z)^2 + c^2}, \quad \text{where} \quad \mathcal{A}_g(z) := \int_0^1 q(t) \sqrt{z^T h_t^{-1} z} dt. \quad (1.2)$$

Moreover, for every primitive $(z, c) \in \mathbb{Z}^{n-1} \times \mathbb{Z}$, the class $r_{z,c}$ is represented by a smooth fibration of $\mathbb{T}^n$ by connected, embedded, calibrated $(n-1)$ -tori. For $z = 0$ these are the horizontal fibers; for $z \neq 0$ they are constant-angle suspensions constructed in Section 3.

The quantity $\mathcal{A}_g(z)$ is the area of the canonical vertical hypersurface obtained by sweeping the flat subtorus dual to z once around the base circle.

The theorem has the following immediate consequence.

Corollary 1.3. *For every $n \geq 3$ there exist smooth nonflat metrics on $\mathbb{T}^n$ for which every primitive class in $H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ admits a foliation by connected, homologically area-minimizing $(n-1)$ -tori. Consequently, the answer to Bangert's question is negative in every dimension $n \geq 3$.*

In his Scalar curvature notes [8], Gromov posed a C^0 -stability question for the rigidity of scalar curvature on $\mathbb{T}^n$: given smooth metrics g_i on $\mathbb{T}^n$ satisfying

$$\text{Sc}(g_i) \geq -\varepsilon_i, \quad \varepsilon_i \rightarrow 0,$$

and converging uniformly to a continuous Riemannian metric g , he asked whether $(\mathbb{T}^n, g)$ must be isometric to a flat torus. His no-gap argument yields, for every primitive codimension-one class, a foliation by g -minimal hypersurfaces, but does not establish that the leaves are totally geodesic. Thus an affirmative answer to Bangert's question would have implied flatness of the limit. Corollary 1.3 shows that the foliation conclusion alone is insufficient. The corresponding C^0 rigidity problem was subsequently resolved by Burkhart-Guim using regularizing Ricci flow [6].

One could ask the following substantially stronger rigidity question: whether the codimension-one stable norm determines the metric. Without further restriction, the answer is negative, as we can increase the metric inside a compact set where no minimizing hypersurface intersects, without changing the set of minimizing hypersurfaces. This is not possible if the underlying metric

is flat: any perturbation of the metric inside a compact set will change the minimizers passing through there. We therefore ask:

Problem 1.4. Let g_0 be a flat metric on the marked torus $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$. If

$$\|\cdot\|_{\text{st},g} = \|\cdot\|_{\text{st},g_0} \quad \text{on } H_{n-1}(\mathbb{T}^n, \mathbb{R}), \quad (1.3)$$

must g itself be flat?

When $n = 2$, this is proved by Bangert in [3]. Our second result shows that when $n = 3$, the codimension-one stable norm alone remains nonrigid even when its entire unit ball is exactly Euclidean.

Theorem 1.5. *There exists a smooth two-parameter family of nonflat metrics*

$$\{g_{\lambda,s} : \lambda > 1, 0 \leq s < 1\}$$

on $\mathbb{T}^3$ such that

$$\|r_{a,b,c}\|_{\text{st},g_{\lambda,s}} = \sqrt{a^2 + b^2 + c^2}$$

for every $(a, b, c) \in H_2(\mathbb{T}^3, \mathbb{R}) \cong \mathbb{R}^3$. Thus every $g_{\lambda,s}$ has exactly the codimension-one stable norm of the unit cubic flat torus. Moreover, for every fixed $\lambda > 1$, the metrics $g_{\lambda,s}$, $0 \leq s < 1$, are pairwise non-isometric.

Therefore, flatness of a metric on $\mathbb{T}^n$ cannot be characterized from the stable norm on H_{n-1} alone, not even locally. For comparison, Osuna proved that, under the unit-volume normalization, equality of both the H_1 and H_{n-1} stable norms with those of a flat torus does force isometry with that flat torus [18]. This leaves open whether equality of the codimension-one stable norm together with equality of volume is already rigid, we formulate this question in Section 5.

The flexibility in our counterexamples is that the minimizing foliation in a mixed class is allowed to depend nonlinearly on that class: its lifted leaves are $z \cdot x + c \sigma_z(t) = \theta$, rather than the standard affine hyperplanes $z \cdot x + ct = \theta$. This distinction is essential. If we strictly require that the foliations on $\mathbb{T}^n$ are all by affine hyperplanes, then we do have flat rigidity. The proof we give in Appendix C follows from a characterization of Hangan [9] on a metric on $\mathbb{R}^n$ for which every hyperplane is minimal.

1.2. Organization of the paper. The paper is organized as follows. Section 2 recalls the definitions of stable norms and calibrations. In Section 3 we prove the exact stable norm formula, construct the calibrated foliations, and derive the counterexamples to Bangert's conjecture. Section 4 constructs the two-parameter family on $\mathbb{T}^3$ with the exact Euclidean stable norm, and we use that family to show that the Euclidean norm plus volume data are not locally injective inside any open neighborhood of the flat metric. Section 5 formulates the remaining stable norm and volume rigidity problem and discusses the higher-dimensional limitations of the Euclidean construction. Appendix A proves the equivalence between Bangert's foliation hypothesis and the differentiability of the stable norm away from the origin. Appendix B proves the stable norm and

volume rigidity statement within the cohomogeneity-one class, while Appendix C establishes rigidity under the stronger affine foliation hypothesis.

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Note added. While preparing this manuscript, we learned of the recent independent work of Marques, Neves, and Sun [14]. Their Theorem 5.1 gives, in an equivalent cofactor formulation, the stable-norm formula for the cohomogeneity-one class considered here and constructs foliations by area-minimizing hypersurfaces in every primitive codimension-one homology class. Their Theorems 1.4 and 1.5 independently provide nonflat examples giving a negative answer to Bangert’s question. They also establish local rigidity near a flat metric under the additional assumption of equal volume. The overlapping results in the two works were obtained independently.

2. STABLE NORM, POINCARÉ DUALITY, AND CALIBRATIONS

Let (M^n, g) be a closed oriented Riemannian manifold. The stable norm of $r \in H_k(M, \mathbb{R})$ can be defined by

$$\|r\|_{\text{st},g} := \inf\{\mathbf{M}_g(T) : T \text{ is a closed real } k\text{-current and } [T] = r\}. \quad (2.1)$$

Equivalently, one may use real Lipschitz cycles. We will only use $k = n - 1$, where minimizers are particularly well suited to calibration methods, see [7] and [1].

A smooth $(n - 1)$ -form Ω has comass at most one if

$$|\Omega(\xi)| \leq 1$$

for every unit simple $(n - 1)$ -vector ξ . If $d\Omega = 0$ and an oriented hypersurface Σ satisfies

$$\Omega|_{\Sigma} = dA_{\Sigma},$$

then Ω calibrates Σ . The standard calibration argument gives the following lemma. See also [10].

Lemma 2.1. *If a closed comass-one $(n - 1)$ -form Ω calibrates an oriented closed hypersurface Σ , then*

$$\mathbf{M}_g(T) \geq \text{Area}_g(\Sigma)$$

for every closed real $(n - 1)$ -current T homologous to Σ . Hence Σ realizes the stable norm of its homology class.

Proof. The comass inequality gives $\mathbf{M}_g(T) \geq T(\Omega)$. Since $d\Omega = 0$ and $[T] = [\Sigma]$, it follows from Stokes’ theorem that

$$T(\Omega) = \Sigma(\Omega) = \text{Area}_g(\Sigma).$$

□

For the torus

$$\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n = \mathbb{T}_x^{n-1} \times S_t^1,$$

we orient by

$$dx^1 \wedge \cdots \wedge dx^{n-1} \wedge dt.$$

Poincaré duality identifies $H_{n-1}(\mathbb{T}^n, \mathbb{R})$ with $H^1(\mathbb{T}^n, \mathbb{R})$. We write $r_{z,c}$ for the class characterized by

$$\text{PD}(r_{z,c}) = \sum_{i=1}^{n-1} z_i [dx^i] + c[dt], \quad (z, c) \in \mathbb{R}^{n-1} \times \mathbb{R}. \quad (2.2)$$

The signs of the coordinate hypersurface basis are chosen consistently with (2.2).

We have the following identity

$$d(\iota_\nu dV_g) = \mathcal{L}_\nu dV_g = (\text{div}_g \nu) dV_g \quad (2.3)$$

for a unit vector field ν . Thus a cooriented hypersurface foliation with divergence-free unit normal is automatically calibrated leafwise.

3. THE EXACT STABLE NORM AND COUNTEREXAMPLES TO BANGERT'S CONJECTURE

In this section, we prove Theorem 1.2 and Corollary 1.3. Recall

$$\mathbb{T}^n = \mathbb{T}_x^{n-1} \times S_t^1$$

with coordinates $(x_1, \dots, x_{n-1}, t)$ and consider metrics

$$g = q(t)^2 dt^2 + h_t, \quad \det h_t = 1, \quad (3.1)$$

where $q : S^1 \rightarrow (0, \infty)$ is smooth and h_t is a smooth periodic family of constant-coefficient metrics on the horizontal torus $\mathbb{T}^{n-1} \times \{t\}$. Geometrically, the metric is a loop of marked flat, unit-volume $(n-1)$ -tori over a circle, the base arc-length element is $q(t)dt$. Its volume form is

$$dV_g = q(t) dx_1 \wedge \cdots \wedge dx_{n-1} \wedge dt. \quad (3.2)$$

The vertical unit field is

$$\nu_0 = q(t)^{-1} \partial_t. \quad (3.3)$$

The horizontal fibers $F_t = \mathbb{T}^{n-1} \times \{t\}$ have mean curvature

$$H_{F_t} = \text{div}_g \nu_0 = \frac{1}{q(t) \sqrt{\det h_t}} \partial_t \left(q(t) \sqrt{\det h_t} \frac{1}{q(t)} \right) \quad (3.4)$$

$$= \frac{1}{q(t)} \frac{d}{dt} \log \sqrt{\det h_t} = 0. \quad (3.5)$$

Thus all horizontal fibers are minimal.

As in (2.2), write

$$\text{PD}(r_{z,c}) = \sum_{i=1}^{n-1} z_i [dx_i] + c[dt], \quad (z, c) \in \mathbb{R}^{n-1} \times \mathbb{R}.$$

We consider two cases: $z = 0$ and $z \neq 0$.

3.1. The horizontal class. When $z = 0$, the primitive classes are $r_{0,\pm 1}$, represented by the horizontal foliation $\{F_t\}_{t \in S^1}$. Since $\operatorname{div}_g \nu_0 = 0$, the form $\iota_{\nu_0} dV_g$ is closed, has comass one, and restricts to the volume form of every F_t . It therefore calibrates the horizontal foliation. Since $\det h_t = 1$, each horizontal fiber has area one, and hence

$$\|r_{0,\pm 1}\|_{\text{st},g} = 1. \quad (3.6)$$

3.2. The classes with nonzero horizontal component. For $z \in \mathbb{R}^{n-1} \setminus \{0\}$, define

$$\mathcal{A}_g(z) := \int_0^1 q(t) \sqrt{z^T h_t^{-1} z} dt. \quad (3.7)$$

Geometrically, for integral $z \neq 0$, $\mathcal{A}_g(z)$ is the area of the hypersurface obtained by sweeping the horizontal subtorus dual to z once around the base circle.

On the universal cover $\mathbb{R}^{n-1} \times \mathbb{R}$, define

$$\tilde{u}_{z,c}(x, t) := z \cdot x + \frac{c}{\mathcal{A}_g(z)} \int_0^t q(s) \sqrt{z^T h_s^{-1} z} ds. \quad (3.8)$$

For $(z, c) \in \mathbb{Z}^{n-1} \times \mathbb{Z}$, it follows from periodicity that for every $(k, \ell) \in \mathbb{Z}^{n-1} \times \mathbb{Z}$ we have

$$\tilde{u}_{z,c}(x + k, t + \ell) - \tilde{u}_{z,c}(x, t) = z \cdot k + c\ell \in \mathbb{Z}.$$

Hence $\tilde{u}_{z,c}$ descends modulo $\mathbb{Z}$ to a circle-valued function

$$u_{z,c} : \mathbb{T}^n \longrightarrow S^1.$$

Since $z \neq 0$, its differential never vanishes, so $u_{z,c}$ is a submersion. Its fibers

$$\Sigma_{z,c,\theta} := u_{z,c}^{-1}(\theta) = \left\{ (x, t) : z \cdot x + \frac{c}{\mathcal{A}_g(z)} \int_0^t q(s) \sqrt{z^T h_s^{-1} z} ds = \theta \pmod{1} \right\} \quad (3.9)$$

belong to the class $r_{z,c}$ and form a smooth foliation of $\mathbb{T}^n$.

Lemma 3.1. *If $(z, c) \in \mathbb{Z}^n$ is primitive and $z \neq 0$, every fiber of $u_{z,c}$ is a connected embedded $(n-1)$ -torus representing $r_{z,c}$.*

Proof. Set

$$\tau_z(t) := \frac{1}{\mathcal{A}_g(z)} \int_0^t q(s) \sqrt{z^T h_s^{-1} z} ds.$$

Then $\tau'_z(t) > 0$ and $\tau_z(t+1) = \tau_z(t) + 1$, so $(x, t) \mapsto (x, \tau_z(t))$ descends to a diffeomorphism of $\mathbb{T}^n$. Under this diffeomorphism, $u_{z,c}$ becomes the linear function

$$(x, \theta) \longmapsto z \cdot x + c\theta \pmod{1}.$$

Its induced map on fundamental groups is

$$\mathbb{Z}^n \longrightarrow \mathbb{Z}, \quad (k, \ell) \longmapsto z \cdot k + c\ell.$$

Primitivity is equivalent to surjectivity, and the kernel is a rank $(n - 1)$ lattice. Hence every fiber is connected and diffeomorphic to $\mathbb{T}^{n-1}$. Moreover,

$$[du_{z,c}] = \sum_{i=1}^{n-1} z_i [dx_i] + c[dt],$$

so an oriented fiber represents $r_{z,c}$. $\square$

Differentiating (3.8) gives

$$du_{z,c} = \sum_{i=1}^{n-1} z_i dx^i + \frac{c}{\mathcal{A}_g(z)} q(t) \sqrt{z^T h_t^{-1} z} dt. \quad (3.10)$$

Its unit normal is then given by

$$\nu_{z,c} = \frac{1}{\sqrt{\mathcal{A}_g(z)^2 + c^2}} \left(\mathcal{A}_g(z) \frac{h_t^{-1} z}{\sqrt{z^T h_t^{-1} z}} + \frac{c}{q(t)} \partial_t \right). \quad (3.11)$$

The horizontal components of $\nu_{z,c}$ depend only on t , whereas

$$q(t) \nu_{z,c}^t = \frac{c}{\sqrt{\mathcal{A}_g(z)^2 + c^2}}$$

is constant. Since $\det h_t = 1$, the volume density of g is $q(t)$, and therefore

$$\operatorname{div}_g \nu_{z,c} = \frac{1}{q(t)} \left[\sum_{i=1}^{n-1} \partial_{x^i} (q(t) \nu_{z,c}^i) + \partial_t (q(t) \nu_{z,c}^t) \right] = 0.$$

Consequently, the closed comass-one form $\iota_{\nu_{z,c}} dV_g$ calibrates every fiber.

It remains only to compute the area of a fiber. Horizontal translations are isometries of g and act transitively on the fibers of $u_{z,c}$, so all fibers have the same area. Since $g^{-1} = q(t)^{-2} \partial_t^2 + h_t^{-1}$, it follows from (3.10) that

$$|du_{z,c}|_g^2 = z^T h_t^{-1} z + q(t)^{-2} \left(\frac{c}{\mathcal{A}_g(z)} q(t) \sqrt{z^T h_t^{-1} z} \right)^2 = z^T h_t^{-1} z \frac{\mathcal{A}_g(z)^2 + c^2}{\mathcal{A}_g(z)^2}.$$

For any $0 \leq \theta_0 \leq 1$, the coarea formula then gives

$$\operatorname{Area}_g(\Sigma_{z,c,\theta_0}) = \int_{S^1} \operatorname{Area}_g(\Sigma_{z,c,\theta}) d\theta \quad (3.12)$$

$$= \int_{\mathbb{T}^n} |du_{z,c}|_g dV_g \quad (3.13)$$

$$= \frac{\sqrt{\mathcal{A}_g(z)^2 + c^2}}{\mathcal{A}_g(z)} \int_0^1 q(t) \sqrt{z^T h_t^{-1} z} dt \quad (3.14)$$

$$= \sqrt{\mathcal{A}_g(z)^2 + c^2}. \quad (3.15)$$

Because the fiber is calibrated, its area is the least mass in its homology class. In particular, we have for every metric in (3.1),

$$\|r_{z,c}\|_{\text{st},g} = \sqrt{\mathcal{A}_g(z)^2 + c^2} \quad (3.16)$$

for every primitive class $r_{z,c} \in H_{n-1}(\mathbb{T}^n, \mathbb{Z})$.

The argument proves the formula (1.2) for primitive integral classes. If $(z, c) = k(z_0, c_0)$ with $k \in \mathbb{N}$ and (z_0, c_0) primitive, then the absolute homogeneity of the stable norm and of $\mathcal{A}_g$ gives

$$\|r_{z,c}\|_{\text{st},g} = k\|r_{z_0,c_0}\|_{\text{st},g} = \sqrt{\mathcal{A}_g(z)^2 + c^2}.$$

Thus the formula holds for every integral class. It then holds for rational classes by rescaling. Finally, rational classes are dense in $H_{n-1}(\mathbb{T}^n, \mathbb{R})$, and both the stable norm and

$$(z, c) \mapsto \sqrt{\mathcal{A}_g(z)^2 + c^2}$$

are continuous. Passing to the limit proves the formula for every real class.

3.3. Counterexamples to Bangert's conjecture. We now derive Corollary 1.3 directly from Theorem 1.2. Let

$$g = q(t)^2 dt^2 + h_t$$

be a metric of the form (3.1), and assume that the periodic family $t \mapsto h_t$ is nonconstant. Theorem 1.2 gives, in every primitive class in $H_{n-1}(\mathbb{T}^n, \mathbb{Z})$, a foliation by connected calibrated homologically area-minimizing tori.

It remains only to verify that g is nonflat. Since the family $t \mapsto h_t$ is nonconstant, the shape operator

$$B(t) = \frac{1}{2q(t)} h_t^{-1} \dot{h}_t$$

of some horizontal slice $F_{t_0} = \mathbb{T}^{n-1} \times \{t_0\}$ is nonzero. The condition $\det h_t \equiv 1$ implies that $B(t_0)$ is trace-free, hence it has principal curvatures $\kappa_+ > 0$ and $\kappa_- < 0$. If E_+ and E_- are corresponding orthonormal principal directions, then the intrinsic flatness of F_{t_0} and the Gauss equation give

$$K_g(E_+, E_-) = -\kappa_+ \kappa_- > 0.$$

Thus g has nonzero sectional curvature and is therefore nonflat. This proves Corollary 1.3.

4. THE EUCLIDEAN STABLE NORM IN DIMENSION THREE

We now specialize the construction of Section 3 to $\mathbb{T}^3 = \mathbb{T}^2 \times S^1$. The aim is to choose a nonconstant unit-volume loop of horizontal metrics for which the swept area norm $\mathcal{A}_g$ is exactly Euclidean. Recall that $g = q(t)^2 dt^2 + h_t$. Since $\det h_t = 1$, it can be orthogonally diagonalized as

$$h_t = R_{\theta(t)}^T \text{diag}(\alpha(t), \beta(t)) R_{\theta(t)},$$

where R_θ is the rotation matrix by angle θ . We try to look for examples where α, β are constant. Set $\alpha(t) = \lambda, \beta(t) = \lambda^{-1}$ where $\lambda > 0$ is a fixed constant and set

$$D_\lambda = \text{diag}(\lambda^2, \lambda^{-2}).$$

For further simplification, let $\theta(t) = 2\pi t$ and set

$$h_\lambda(t) = R_{2\pi t}^T D_\lambda R_{2\pi t}. \quad (4.1)$$

We then have that $\det h_\lambda(t) = 1$.

For $z \in \mathbb{R}^2$, the function

$$z \mapsto \int_0^1 \sqrt{z^T h_\lambda(t)^{-1} z} dt$$

is invariant under planar rotations and is positively homogeneous. Consequently, it is a constant multiple of the Euclidean norm. We write

$$\int_0^1 \sqrt{z^T h_\lambda(t)^{-1} z} dt = \kappa_\lambda |z|, \quad (4.2)$$

where evaluation at $z = e_1$ gives

$$\kappa_\lambda = \int_0^1 \sqrt{\lambda^{-2} \cos^2(2\pi t) + \lambda^2 \sin^2(2\pi t)} dt. \quad (4.3)$$

The next theorem gives a general condition on the vertical function $q(t)$ under which the codimension-one stable norm is Euclidean.

Theorem 4.1. *Fix $\lambda > 1$, and let $q : S^1 \rightarrow (0, \infty)$ be a smooth function satisfying*

$$q(t) + q\left(t + \frac{1}{2}\right) = \frac{2}{\kappa_\lambda} \quad \text{for every } t \in S^1. \quad (4.4)$$

Then the metric

$$g_{\lambda,q} = q(t)^2 dt^2 + h_\lambda(t) \quad (4.5)$$

on $\mathbb{T}^3$ is nonflat and satisfies

$$\|r_{a,b,c}\|_{\text{st}, g_{\lambda,q}} = \sqrt{a^2 + b^2 + c^2} \quad (4.6)$$

for every $(a, b, c) \in H_2(\mathbb{T}^3, \mathbb{R}) \cong \mathbb{R}^3$. Moreover,

$$\text{Vol}(\mathbb{T}^3, g_{\lambda,q}) = \kappa_\lambda^{-1}. \quad (4.7)$$

Proof. It follows from the definition that h_λ satisfies the periodicity condition:

$$h_\lambda\left(t + \frac{1}{2}\right) = h_\lambda(t)$$

For $z \in \mathbb{R}^2$, we compute the function $\mathcal{A}_{g_{\lambda,q}}$ as in Section 3, using a change of variable as follows:

$$\begin{aligned} \mathcal{A}_{g_{\lambda,q}}(z) &= \int_0^1 q(t) \sqrt{z^T h_{\lambda}(t)^{-1} z} dt \\ &= \int_0^{1/2} \left[q(t) + q\left(t + \frac{1}{2}\right) \right] \sqrt{z^T h_{\lambda}(t)^{-1} z} dt \\ &= \frac{2}{\kappa_{\lambda}} \int_0^{1/2} \sqrt{z^T h_{\lambda}(t)^{-1} z} dt \\ &= \frac{1}{\kappa_{\lambda}} \int_0^1 \sqrt{z^T h_{\lambda}(t)^{-1} z} dt = |z|, \end{aligned}$$

where the last equality follows from (4.2). Theorem 1.2 therefore gives (4.6).

Since $\det h_{\lambda}(t) = 1$, the volume form on $\mathbb{T}^3$ is $dV_g = q(t) dx dy dt$. Integrating (4.4) over $[0, \frac{1}{2}]$ with a change of variable gives

$$\int_0^1 q(t) dt = \kappa_{\lambda}^{-1},$$

which proves (4.7). Finally, the map $t \mapsto h_{\lambda}(t)$ is non-constant when $\lambda > 1$, so the nonflatness of the constructed metrics follows from the argument in Section 3. $\square$

Remark 4.2. Condition (4.4) is equivalent to

$$q(t) = \kappa_{\lambda}^{-1}(1 + \psi(t)), \quad \psi\left(t + \frac{1}{2}\right) = -\psi(t), \quad \|\psi\|_{C^0} < 1.$$

The space of such smooth functions ψ is an infinite dimensional subspace of $C^{\infty}(S^1)$.

By specializing $q(t)$ to some special form, we can state and prove Theorem 1.5 in a more precise form as follows.

Corollary 4.3. *For $(\lambda, s) \in (1, \infty) \times [0, 1)$, define*

$$q_{\lambda,s}(t) = \kappa_{\lambda}^{-1}(1 + s \sin(2\pi t)) \tag{4.8}$$

and

$$g_{\lambda,s} = q_{\lambda,s}(t)^2 dt^2 + h_{\lambda}(t). \tag{4.9}$$

Then every $g_{\lambda,s}$ is nonflat and has the Euclidean codimension-one stable norm,

$$\|r_{a,b,c}\|_{\text{st}, g_{\lambda,s}} = \sqrt{a^2 + b^2 + c^2}. \tag{4.10}$$

For each fixed $\lambda > 1$, the metrics $g_{\lambda,s}$ have the same volume κ_{λ}^{-1} and are pairwise non-isometric. For each fixed $s < 1$, as $\lambda \downarrow 1$ they converge, after pullback by diffeomorphisms isotopic to the identity, to the unit Euclidean flat metric in the C^{∞} topology.

Proof. The function $q(t) = \kappa_\lambda^{-1}(1 + s \sin(2\pi t))$ obviously satisfies the condition (4.4); therefore the stable-norm, volume, and nonflatness statements follow from Theorem 4.1.

It remains to distinguish the metrics for different values of s . We can do that by directly computing their scalar curvatures.

The shape operator of the horizontal two-tori can be computed as:

$$B_{\lambda,s}(t) = \frac{1}{2q_{\lambda,s}} h_\lambda(t)^{-1} \dot{h}_\lambda(t).$$

Using the scalar Gauss-Riccati identity $\text{Sc}(g_{\lambda,s}) = -|B_{\lambda,s}|^2$, one obtains

$$\text{Sc}(g_{\lambda,s})(t) = -\frac{2\pi^2 \kappa_\lambda^2 (\lambda^2 - \lambda^{-2})^2}{(1 + s \sin(2\pi t))^2}. \quad (4.11)$$

Taking the infimum over t we have

$$\min \text{Sc}(g_{\lambda,s}) = -\frac{2\pi^2 \kappa_\lambda^2 (\lambda^2 - \lambda^{-2})^2}{(1 - s)^2}.$$

It follows that $g_{\lambda,s}$ and $g_{\lambda,s'}$ are not isometric when $s \neq s'$ and $\lambda \neq 1$. Metrics corresponding to two different values $\lambda, \lambda' > 1$ are distinguished by their volumes.

Finally,

$$h_\lambda(t) \longrightarrow I, \quad \kappa_\lambda \longrightarrow 1$$

in C^∞ as $\lambda \downarrow 1$. Therefore

$$g_{\lambda,s} \longrightarrow (1 + s \sin(2\pi t))^2 dt^2 + dx^2 + dy^2.$$

The limiting metric is the pullback of the unit Euclidean metric under the circle reparametrization

$$\tau_s(t) = \int_0^t (1 + s \sin(2\pi u)) du.$$

Since this reparametrization is isotopic to the identity, the last conclusion follows. This completes the proof. $\square$

5. OPEN QUESTIONS

5.1. Stable norm and volume rigidity on $\mathbb{T}^n$. Theorem 1.5 shows that the codimension-one stable norm alone does not determine flatness, even when it agrees exactly with the stable norm of the unit Euclidean flat torus. One could ask for an additional geometric condition that could characterize flatness. Motivated by Osuna's result in [18] we ask the following sharper rigidity question.

Question 5.1. *Let g be a smooth Riemannian metric on the standard marked torus $\mathbb{T}^n$. Suppose that*

$$\|r_{a_1, a_2, \dots, a_n}\|_{\text{st}, g} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$

for every $(a_1, \dots, a_n) \in H_{n-1}(\mathbb{T}^n, \mathbb{R}) \cong \mathbb{R}^n$, and that

$$\text{Vol}(\mathbb{T}^n, g) = 1.$$

Must g be isometric to a flat metric by a diffeomorphism isotopic to the identity?

For $n = 3$, the metrics constructed in Section 4, however, satisfy

$$\text{Vol}(\mathbb{T}^3, g_{\lambda,s}) = \kappa_\lambda^{-1} < 1 \quad (\lambda > 1),$$

so they do not answer this question negatively. On the other hand, for each fixed $\lambda > 1$, each of the metrics $g_{\lambda,s}$ is not even locally rigid with respect to the data (Euclidean stable norm, volume). In [14], the authors prove that the flat metric is locally rigid under this data.

Within the class of metric in Theorem 1.2, the question has a sharp affirmative answer. We give a proof in Proposition B.1 in Appendix B.

5.2. Higher-dimensional prescribed Euclidean stable norm. A naive generalization of the construction in Section 4 does not work. The same rotation trick fails to yield $\mathcal{A}_g(z) = |z|$ for $\mathbb{R}^3$.

Question 5.2. For $n \geq 4$, does there exist a smooth nonflat metric on the standard torus $\mathbb{T}^n$ whose stable norm on $H_{n-1}(\mathbb{T}^n, \mathbb{R})$ is exactly

$$(a_1, \dots, a_n) \mapsto \sqrt{a_1^2 + \dots + a_n^2}$$

under the standard integral identification?

APPENDIX A. MINIMIZING FOLIATIONS AND DIFFERENTIABILITY OF THE STABLE NORM

We record the precise relation between Bangert's foliation condition and the differentiability theory of the codimension-one stable norm. The result is a torus consequence of Auer-Bangert's differentiability theorem [1], Junginger-Gestrich's gap theorem [13], and a compactness argument that passes from rational foliations to irrational directions.

Let $\Gamma = H_1(\mathbb{T}^n, \mathbb{Z})$, and let $I : \Gamma \times H_{n-1}(\mathbb{T}^n, \mathbb{R}) \rightarrow \mathbb{R}$ be the intersection pairing. For $0 \neq \alpha \in H_{n-1}(\mathbb{T}^n, \mathbb{R})$ set

$$K(\alpha) = \{k \in \Gamma : I(k, \alpha) = 0\} \tag{A.1}$$

and

$$V(\alpha) = \{\beta : I(k, \beta) = 0 \text{ for every } k \in K(\alpha)\}. \tag{A.2}$$

The space $V(\alpha)$ is the smallest linear subspace containing α and spanned by integral classes.

We briefly recall the minimizing-lamination notation used in [13]. On the Abelian cover $\mathbb{R}^n \rightarrow \mathbb{T}^n$, let $\mathcal{M}$ denote the nonzero homologically minimizing codimension-one boundaries with the Birkhoff property. Every $R \in \mathcal{M}$ has a normalized first invariant $\alpha_1(R)$ satisfying $\|\alpha_1(R)\|_g = 1$, followed possibly by secondary invariants. For $\|\alpha\|_g = 1$, the set $\mathcal{M}(\alpha)$ consists of the currents whose first invariant is α and which have no secondary invariant; equivalently,

they are invariant under every deck transformation in $K(\alpha)$. The set $\mathcal{M}(\alpha)$ is totally ordered and gives a minimal lamination of $\mathbb{R}^n$.

We use the following four facts from the cited works. First, Auer and Bangert [1] prove that $\|\cdot\|_g|_{V(\alpha)}$ is differentiable at α . Second, if α is rationally dependent, Junginger-Gestrich [13] proves that $\|\cdot\|_g$ is differentiable at α in the remaining directions if and only if $\mathcal{M}(\alpha)$ has no gaps. Third, on a torus every element of $\mathcal{M}$ has connected support (see [17]). Finally, the subfamily of minimizing currents whose supports meet a fixed compact set is compact, and the first invariant is continuous under this convergence.

Theorem A.1. *Let g be a smooth Riemannian metric on $\mathbb{T}^n$. The following statements are equivalent.*

- (i) *Every primitive class in $H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ admits a foliation by connected homologically area-minimizing hypersurfaces.*
- (ii) *The stable norm $\|\cdot\|_{\text{st},g}$ is differentiable at every nonzero point of $H_{n-1}(\mathbb{T}^n, \mathbb{R})$.*

In arbitrary dimension, statement (i) may be read in the codimension-one minimizing current sense. For $n \leq 7$ the leaves are smooth embedded hypersurfaces.

Proof. Assume first that (i) holds. Fix $\alpha \neq 0$ and normalize it by $\|\alpha\|_g = 1$.

If α is totally irrational, then $V(\alpha) = H_{n-1}(\mathbb{T}^n, \mathbb{R})$, and the full differentiability follows directly from Auer-Bangert [1]. If the direction of α is rational, let r be the primitive integral class on its ray. The assumed foliation in the class r lifts to a foliation of $\mathbb{R}^n$ by elements of $\mathcal{M}(\alpha)$. Hence $\mathcal{M}(\alpha)$ has no gaps, and the Auer-Bangert-Junginger-Gestrich criterion gives differentiability of $\|\cdot\|_g$ at α .

It remains to consider the rationally dependent case where

$$1 < \dim V(\alpha) < n.$$

The group $V(\alpha) \cap H_{n-1}(\mathbb{T}^n, \mathbb{Z})$ is a full, saturated lattice in $V(\alpha)$, and primitive lattice directions are dense in the projective space of $V(\alpha)$. We may therefore choose primitive integral classes $r_j \in V(\alpha)$ such that

$$\alpha_j := \frac{r_j}{\|r_j\|_g} \longrightarrow \alpha. \quad (\text{A.3})$$

By (A.2),

$$K(\alpha) \subset K(r_j) = K(\alpha_j) \quad \text{for every } j. \quad (\text{A.4})$$

Fix an arbitrary point $x \in \mathbb{R}^n$. A foliation of $\mathbb{T}^n$ by connected closed hypersurfaces in a primitive class is a fibration over S^1 whose cohomology class is Poincaré dual to that homology class. On the universal cover, its leaf parameter changes under a deck transformation k by $I(k, r_j)$. Hence every $k \in K(r_j)$ preserves each lifted leaf. Let R_j^x be the lifted leaf through x , with the orientation for which its first invariant is α_j . It is a homologically minimizing Birkhoff current and is invariant under every deck transformation in $K(r_j)$. In particular,

$$(\tau_k)_\# R_j^x = R_j^x \quad \text{for every } k \in K(\alpha). \quad (\text{A.5})$$

The compactness theorem for minimizing Birkhoff currents meeting the compact set $\{x\}$ gives, after passage to a subsequence,

$$R_j^x \longrightarrow R^x \in \mathcal{M}, \quad x \in \text{spt } R^x. \quad (\text{A.6})$$

Continuity of the first invariant and (A.3) imply

$$\alpha_1(R^x) = \alpha.$$

For every $k \in K(\alpha)$, equation (A.5) passes to the limit, so $(\tau_k)_\# R^x = R^x$. The classification of Birkhoff minimizers then shows that R^x has no secondary invariant, hence

$$R^x \in \mathcal{M}(\alpha).$$

Since x was arbitrary, the supports of the currents in $\mathcal{M}(\alpha)$ cover $\mathbb{R}^n$. The set $\mathcal{M}(\alpha)$ is already a lamination, so it is a foliation. Junginger-Gestrich's gap criterion gives differentiability in all directions outside $V(\alpha)$, while Auer-Bangert gives differentiability in the directions of $V(\alpha)$. Thus $\|\cdot\|_g$ is differentiable at α . This proves (ii).

Conversely, assume (ii) and let r be a primitive integral class. Set $\alpha = r/\|r\|_g$. This is a rational direction, and full differentiability at α implies, by the gap criterion, that $\mathcal{M}(\alpha)$ foliates $\mathbb{R}^n$. Every leaf is invariant under $K(r)$, and the quotient by the deck group projects the lamination to a foliation of $\mathbb{T}^n$ by closed homologically area-minimizing hypersurfaces in the class r . This proves (i). $\square$

APPENDIX B. STABLE NORM AND VOLUME RIGIDITY IN THE COHOMOGENEITY-ONE CLASS

We record the sharp form of Question 5.1 within the metric class used in the main construction.

Proposition B.1. *Let*

$$g = q(t)^2 dt^2 + h_t, \quad \det h_t = 1,$$

be a metric on $\mathbb{T}^3 = \mathbb{T}^2 \times S^1$. If its stable norm on $H_2(\mathbb{T}^3, \mathbb{R})$ is the Euclidean norm, then

$$\text{Vol}(\mathbb{T}^3, g) \leq 1.$$

Equality holds if and only if g is isometric to the unit Euclidean flat metric by a diffeomorphism isotopic to the identity.

Proof. By Theorem 1.2, the Euclidean stable-norm hypothesis is equivalent to

$$\mathcal{A}_g(z) = \int_0^1 q(t) \sqrt{z^T h_t^{-1} z} dt = |z| \quad \text{for every } z \in \mathbb{R}^2. \quad (\text{B.1})$$

For each t , let

$$E_t = \{v \in \mathbb{R}^2 : h_t(v, v) \leq 1\}$$

be the horizontal unit ellipse. Since $\det h_t = 1$, the Euclidean area of E_t is π , and its support function is

$$h_{E_t}(z) = \sqrt{z^T h_t^{-1} z}.$$

Average (B.1) over the Euclidean unit circle. By Fubini's theorem and Cauchy's perimeter formula, we have

$$\begin{aligned} 1 &= \frac{1}{2\pi} \int_0^{2\pi} \mathcal{A}_g((\cos \theta, \sin \theta)) d\theta \\ &= \int_0^1 q(t) \frac{\text{Per}(E_t)}{2\pi} dt. \end{aligned}$$

The planar isoperimetric inequality gives $\text{Per}(E_t) \geq 2\pi$, with equality precisely when E_t is the unit disk. Therefore

$$1 \geq \int_0^1 q(t) dt = \text{Vol}(\mathbb{T}^3, g).$$

If equality holds, then $q(t) > 0$ forces $\text{Per}(E_t) = 2\pi$ for every t . Hence E_t is the unit disk and $h_t \equiv I$. The metric is consequently

$$g = q(t)^2 dt^2 + dx^2 + dy^2.$$

Since $\int_0^1 q(t) dt = 1$, the function

$$\tau(t) = \int_0^t q(u) du$$

defines an orientation-preserving circle diffeomorphism isotopic to the identity, and in the coordinates (x, y, τ) one has

$$g = dx^2 + dy^2 + d\tau^2.$$

Conversely, the unit Euclidean flat metric clearly realizes equality. This completes the proof. $\square$

APPENDIX C. RIGIDITY OF THE AFFINE FOLIATION CONDITION

We now fix the standard affine structure on $\mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$. For a primitive vector $p \in \mathbb{Z}^n$, let

$$\mathcal{F}_p := \left\{ \Sigma_{p,s} : p \cdot x = s \pmod{1} \right\}_{s \in S^1} \quad (\text{C.1})$$

be the standard affine foliation in the class Poincaré dual to p .

Theorem C.1. *Let $n \geq 2$, and let g be a smooth Riemannian metric on the standard affine torus $\mathbb{T}^n$. Suppose that for every primitive covector $p \in \mathbb{Z}^n$, every leaf of the affine foliation $\mathcal{F}_p$ is g -minimal. Then the coefficients of g are constant in the standard affine coordinates. In particular, g is flat.*

The proof combines Hangan's local classification of metrics with minimal affine hyperplanes in [9] with a global periodicity argument. Hangan first derived the differential system imposed by the vanishing of the mean curvature of all affine hyperplanes and then integrated it in terms of Euclidean Killing 2-tensors.

Proof. Lift g to a $\mathbb{Z}^n$ -periodic metric (still denoted by g) on $\mathbb{R}^n$. For a nonzero constant vector $p \in \mathbb{R}^n$, let

$$u_p(x) = p \cdot x.$$

By the assumption, using compactness property of stable minimal hypersurfaces, every affine hyperplane is minimal. The mean curvature of the level hyperplane through x , computed with the unit normal $\nabla^g u_p / |\nabla^g u_p|_g$, is

$$H_g(x, p) = \operatorname{div}_g \left(\frac{\nabla^g u_p}{|\nabla^g u_p|_g} \right) = 0.$$

Write

$$g = g_{ij}(x) dx^i dx^j, \quad |g| := \det(g_{ij}).$$

Hangan's correspondence now applies. In the notation of [9], the positive-definite symmetric tensor

$$G := |g|^{-\frac{2}{n+1}} g \tag{C.2}$$

satisfies the Euclidean Killing-tensor equation

$$\partial_i G_{jk} + \partial_j G_{ki} + \partial_k G_{ij} = 0. \tag{C.3}$$

The Killing equation has a direct dynamical meaning. For every constant vector $v \in \mathbb{R}^n$ and every Euclidean straight line $\gamma(t) = x + tv$, equation (C.3) gives

$$\frac{d}{dt} G_{x+tv}(v, v) = (\partial_v G)_{x+tv}(v, v) = 0. \tag{C.4}$$

Thus $G(v, v)$ is a quadratic first integral of the standard straight-line flow.

Choose v whose coordinates are rationally independent. The projected line $x + tv \pmod{\mathbb{Z}^n}$ is dense in $\mathbb{T}^n$. By (C.4), the continuous function $G_x(v, v)$ is constant along this dense orbit and hence constant on all of $\mathbb{T}^n$. Since totally irrational vectors are dense in $\mathbb{R}^n$, we have that for fixed $x, y \in \mathbb{T}^n$, the quadratic polynomial

$$v \mapsto G_x(v, v) - G_y(v, v)$$

vanishes on a dense subset of $\mathbb{R}^n$ and hence vanishes identically. Polarization gives

$$G_x(v, w) = G_y(v, w) \quad \text{for all } x, y \in \mathbb{T}^n \text{ and } v, w \in \mathbb{R}^n.$$

Consequently, G has constant coefficients. This implies that $g = |G|^{\frac{2}{1-n}} G$ also has constant coefficients and is flat. This completes the proof. $\square$

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